BITQUBE

B T Q

A Decentralized Layer-1 Proof-of-Work Blockchain

Secured by the KawPoW Mining Algorithm

WHITE PAPER

Version 1.0

www.bitqube.org

Legal Disclaimer

This white paper is intended to provide information about the BitQube blockchain, its technical design, and its intended ecosystem. It is not financial, investment, legal, or tax advice, and it does not constitute an offer or solicitation to buy or sell any asset.

Blockchain technology and digital assets involve technical, operational, and market risks. Readers should conduct their own independent research before participating in the BitQube network or related activities.

Any roadmap items, future features, integrations, or ecosystem plans described in this document reflect current project objectives and may change as development progresses.

BTQ is not marketed as a security, ownership interest, profit-sharing instrument, or guaranteed investment product. No guarantee is made regarding future value, market performance, liquidity, or financial returns.

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1. Executive Summary

BitQube (BTQ) is an independent Layer-1 Proof-of-Work blockchain secured by the KawPoW mining algorithm. The network is designed to provide an open, decentralized blockchain where GPU miners validate transactions and secure the network through distributed consensus.

BTQ is the native coin of the BitQube blockchain. New BTQ is introduced through mining according to the network's emission schedule, with every valid block contributing to the security and continuity of the blockchain.

The BitQube project aims to provide transparent blockchain infrastructure that encourages community participation, decentralized mining, and long-term network resilience.

2. Vision & Mission

Vision

To develop an open, secure, and community-driven Layer-1 blockchain that enables decentralized participation through GPU mining and serves as a reliable foundation for the BitQube ecosystem.

Mission

- Maintain an independent Layer-1 blockchain.
- Secure the network using the KawPoW Proof-of-Work consensus mechanism.
- Encourage broad participation by miners and node operators.
- Provide transparent blockchain infrastructure.
- Support long-term ecosystem growth through open technologies and community collaboration.

3. Introduction

Blockchain networks provide a decentralized method of recording and validating transactions without relying on a central authority. In a Proof-of-Work system, miners contribute computational power to validate transactions, create new blocks, and secure the network.

BitQube is built as an independent Layer-1 blockchain using the KawPoW mining algorithm. By leveraging GPU mining, the network aims to promote distributed participation and robust network security while remaining accessible to a broad mining community.

As an independent blockchain, BitQube maintains its own ledger, consensus mechanism, block production, and native coin (BTQ). The network operates separately from other blockchains while supporting future interoperability where appropriate.

4. The Need for Decentralized Proof-of-Work

Proof-of-Work has demonstrated the ability to secure decentralized networks by requiring computational effort to validate transactions and produce blocks.

BitQube adopts the KawPoW algorithm to support GPU-based mining and encourage distributed participation. A decentralized mining community can help strengthen network resilience by reducing reliance on a small number of participants.

The project's objective is to maintain a transparent and verifiable blockchain where anyone meeting the technical requirements can participate in securing the network.

5. BitQube Overview

BitQube is an independent Layer-1 blockchain with the following characteristics:

- Native coin: BTQ
- Proof-of-Work consensus
- KawPoW mining algorithm
- GPU-minable network
- Community participation
- Independent blockchain ledger
- Decentralized block validation
- Open blockchain infrastructure

The BitQube blockchain is intended to provide a secure and transparent foundation for users, miners, developers, and future ecosystem services.

6. Blockchain Architecture

BitQube is an independent Layer-1 Proof-of-Work blockchain with its own ledger, peer-to-peer network, consensus mechanism, and native coin (BTQ). The network is maintained by distributed participants who validate transactions, produce blocks, and help secure the blockchain.

Core Components

Blockchain Ledger

- Immutable record of confirmed transactions.
- Sequential chain of cryptographically linked blocks.

Peer-to-Peer Network

- Nodes communicate directly without a central authority.
- Transactions and blocks are propagated across the network.

Mining Nodes

- GPU miners perform Proof-of-Work calculations using the KawPoW algorithm.
- Successful miners produce valid blocks and receive block rewards according to the network rules.

Full Nodes

- Independently verify transactions and blocks.
- Maintain a complete copy of the blockchain.
- Help strengthen network decentralization and resilience.

7. Proof-of-Work (KawPoW)

BitQube uses the KawPoW Proof-of-Work algorithm to secure the blockchain.

In a Proof-of-Work system, miners compete to solve computational challenges. The first miner to produce a valid solution creates the next block, which is independently verified by the network before being added to the blockchain.

Objectives

- Secure the blockchain.
- Enable decentralized block production.
- Support GPU mining.
- Encourage broad community participation.
- Protect the integrity of the distributed ledger.

The network's security increases with greater participation from miners and node operators.

8. Blocknomics

Definition

Blocknomics is the economic model of the BitQube blockchain. It defines how BTQ is created, distributed, and sustained through Proof-of-Work mining and network participation.

Components of Blocknomics

Maximum Supply

The total maximum supply of BTQ is fixed at 8,000,000 BTQ, providing a predictable issuance model.

Initial Allocation

A predefined allocation of 641,600 BTQ (8.02% of maximum supply) is reserved as the Ecosystem Reserve for development, infrastructure, security, partnerships, exchange listings, and long-term sustainability, as defined by the network's launch parameters.

Mining Distribution

The remaining 7,358,400 BTQ (91.98%) is distributed through Proof-of-Work mining over time according to the network's emission schedule.

Block Rewards

Miners who successfully produce valid blocks receive block rewards determined by the protocol. These rewards decrease across eight defined phases according to the predefined emission schedule (see Section 22).

Transaction Fees

Users pay network transaction fees to have transactions included in blocks. These fees contribute to the network's operation and provide an additional incentive for block producers, particularly after mining emission concludes.

Economic Sustainability

BitQube's economic model is designed to balance:

- Predictable coin issuance.
- Incentives for miners.
- Long-term network participation.
- Transparent monetary policy.

9. Network Security

BitQube's security is based on decentralized Proof-of-Work consensus.

Security Principles

- Independent block verification.
- Distributed mining participation.
- Cryptographic block validation.
- Immutable transaction history.
- Open-source verification.

Miner Participation

As additional miners contribute hash rate, the network becomes more resistant to attacks that depend on controlling a large share of the total computational power.

Node Verification

Every full node independently validates blocks and transactions according to the protocol rules before accepting them into the blockchain. This distributed verification process helps maintain network consistency without relying on a central authority.

10. Mining Ecosystem

Mining is the foundation of the BitQube network. GPU miners perform Proof-of-Work calculations, validate transactions, produce blocks, and secure the blockchain.

Mining Objectives

- Secure the network.
- Produce new blocks.
- Distribute newly issued BTQ according to the protocol.
- Maintain decentralization through community participation.

Mining Participation

Participants may choose to:

- Mine independently.
- Join compatible mining pools.

- Operate full nodes in support of the network.

The network benefits from broad and geographically distributed participation, which can improve resilience and decentralization.

11. Nodes & Network Infrastructure

The BitQube network is supported by a decentralized infrastructure of full nodes, mining nodes, wallets, blockchain explorers, and mining pools.

Full Nodes

Full nodes are responsible for:

- Maintaining a complete copy of the blockchain.
- Verifying every transaction and block.
- Enforcing consensus rules.
- Broadcasting validated blocks to the network.

Anyone meeting the hardware and network requirements may operate a full node, contributing to the decentralization and resilience of the BitQube blockchain.

Mining Nodes

Mining nodes perform Proof-of-Work calculations using the KawPoW algorithm. When a valid block is found, it is broadcast to the network for independent verification.

Mining Pools

Mining pools allow participants to combine computational resources and share rewards according to the pool's payout methodology. Pools can reduce reward variance for individual miners while still contributing to network security.

Blockchain Explorer

A blockchain explorer provides transparent access to on-chain information, allowing users to view:

- Block height
- Block details
- Transactions
- Wallet addresses
- Network statistics
- Mining information

This transparency enables anyone to independently verify blockchain activity.

12. Wallet Infrastructure

BTQ is managed using wallets that allow users to control their own private keys.

Wallet functionality may include:

- Sending and receiving BTQ
- Viewing balances
- Transaction history
- Address generation
- Backup and recovery
- Private key management

Users are responsible for securely storing their wallet backups, recovery phrases, and private keys. Loss of these credentials may result in permanent loss of access to funds.

13. BitQube Utility

BTQ functions as the native coin of the BitQube blockchain.

Potential uses within the ecosystem may include:

- Peer-to-peer value transfer.
- Payment for products and services offered by participating businesses.
- Settlement within supported ecosystem applications.
- Community-driven transactions.
- Other utilities adopted by ecosystem participants over time.

Adoption depends on ecosystem growth, community participation, and third-party integration.

14. Ecosystem Integration

BitQube is intended to serve as a foundational blockchain within the broader BitQube ecosystem.

Examples of planned or developing ecosystem components include:

- Non-custodial wallet applications.
- Payment integrations.
- Mining-related infrastructure.
- Community services.
- Additional applications developed by ecosystem participants.

Each component may evolve independently while interacting with the BitQube blockchain according to its technical design.

15. Technical Specifications

The following table summarizes the core protocol identity of BitQube.

Specification	Value
Blockchain Type	Layer-1
Consensus	Proof-of-Work
Mining Algorithm	KawPoW
Native Coin	BTQ
Ledger	Public
Network Model	Peer-to-Peer
Block Validation	Decentralized
Source Code	Public repository (if applicable)

16. Development Roadmap

The roadmap outlines the project's intended direction. Actual milestones may change based on development priorities, community feedback, and technical considerations.

Phase 1 — Network Launch

- Mainnet launch
- Open GPU mining
- Mining pool deployment
- Block explorer
- Desktop wallet release
- Community onboarding

Phase 2 — Ecosystem Development

- Mobile wallet
- Infrastructure improvements
- GPU cloud mining platform
- Merchant adoption
- Community expansion
- Documentation improvements

Phase 3 — Market Expansion

- Decentralized trading infrastructure
- Additional wallet integrations
- Exchange listings
- Developer ecosystem growth
- Public API enhancements

Phase 4 — Enterprise Adoption

- Business partnerships
- Infrastructure services
- Payment integrations
- Global community expansion
- Long-term protocol maintenance

17. Risks & Limitations

Participation in blockchain networks involves various risks, including:

- Software defects
- Cybersecurity threats
- Market volatility
- Regulatory changes
- Infrastructure failures
- User errors, such as loss of private keys

Users should understand these risks and take appropriate precautions.

18. Conclusion

BitQube is an independent Layer-1 Proof-of-Work blockchain secured by the KawPoW algorithm. Its objective is to provide an open, decentralized, and transparent blockchain where miners, node operators, developers, and users can participate in maintaining the network.

By combining decentralized consensus, GPU mining, and community participation, BitQube seeks to provide a sustainable foundation for long-term blockchain development.

The success of the network ultimately depends on continued participation, responsible development, and collaboration among its community members.

19. Technical Appendix

Appendix A — Network Design Principles

The BitQube blockchain is designed around the following principles:

Decentralization

BitQube operates as an independent peer-to-peer blockchain. No single participant controls block production, transaction validation, or network operation.

Transparency

All confirmed transactions and blocks are recorded on the public blockchain and can be independently verified using compatible blockchain explorers and full nodes.

Security

Network security is provided through the KawPoW Proof-of-Work consensus mechanism and the collective computational power contributed by miners.

Open Participation

Anyone meeting the technical requirements may participate by running a full node or mining node, subject to applicable laws and regulations.

Appendix B — Network Lifecycle

The BitQube blockchain processes transactions through the following stages:

- A user creates and signs a transaction.
- The transaction is broadcast to the peer-to-peer network.
- Nodes validate the transaction according to protocol rules.
- Valid transactions enter the memory pool.
- Miners select transactions and construct a candidate block.
- Miners perform KawPoW computations.
- A valid block is broadcast to the network.
- Full nodes independently verify the block.
- Once accepted, the block becomes part of the blockchain.
- The process repeats for subsequent blocks.

Appendix C — Community Participation

Community participation is an important aspect of the BitQube network. Participants may contribute by:

- Mining BTQ.
- Operating full nodes.
- Developing applications.
- Reporting software issues.
- Improving documentation.
- Supporting community education.
- Contributing open-source improvements.

Appendix D — Ecosystem Philosophy

BitQube aims to encourage an ecosystem where blockchain infrastructure can support practical applications developed by independent contributors and organizations. Potential areas of development include:

- Digital payments.
- Mining infrastructure.
- Wallet software.
- Developer tools.
- Business integrations.
- Community applications.

Appendix E — Future Development

Future development priorities may include:

- Network optimization.
- Wallet enhancements.
- Improved node software.
- Enhanced developer documentation.
- Additional ecosystem services.
- Security improvements.
- Community growth initiatives.

20. Protocol Specifications

This chapter summarizes the core protocol parameters of the BitQube blockchain. The values listed below should match the official software implementation.

Parameter	Value
Blockchain	BitQube
Native Coin	BTQ
Blockchain Type	Layer-1
Consensus	Proof-of-Work
Mining Algorithm	KawPoW
Maximum Supply	8,000,000 BTQ
Premine	641,600 BTQ
Mineable Supply	7,358,400 BTQ
Block Time	Protocol Value
Initial Block Reward	Protocol Value
Current Block Reward	Protocol Value
Reward Adjustment	Protocol Value
Difficulty Adjustment	Protocol Value
Transaction Confirmation	Protocol Value
Address Format	Protocol Value
P2P Port	Protocol Value
RPC Port	Protocol Value

21. Consensus Rules

BitQube uses the KawPoW Proof-of-Work consensus algorithm.

The network operates according to the following principles:

- Every block must satisfy the Proof-of-Work requirements defined by the protocol.
- Nodes independently verify all transactions and blocks before accepting them.
- Invalid blocks are rejected.

- The valid blockchain with the greatest cumulative Proof-of-Work is recognized by participating nodes according to the protocol rules.
- Consensus is achieved without reliance on a central authority.

These rules help maintain consistency across the distributed network.

22. Emission Schedule

BTQ is issued exclusively according to the blockchain protocol. The total supply is permanently limited to 8,000,000 BTQ.

Initial Allocation

Allocation	BTQ
Premine	641,600
Mineable Supply	7,358,400
Total Supply	8,000,000

The protocol defines the issuance of new BTQ through block rewards until the maximum supply is reached.

Emission Model

The emission schedule is determined by the blockchain protocol.

Phase	Block Reward	Coins Issued
Phase 1	Protocol Value	Protocol Value
Phase 2	Protocol Value	Protocol Value
Phase 3	Protocol Value	Protocol Value
Phase 4	Protocol Value	Protocol Value
Final Phase	Protocol Value	Remaining Supply

The protocol software is the authoritative source for block reward calculations.

23. Supply & Distribution

The distribution of BTQ follows predefined protocol rules.

Category	Amount	Percentage
Premine (Ecosystem Reserve)	641,600 BTQ	8.02%
Community Mining	7,358,400 BTQ	91.98%
Total Supply	8,000,000 BTQ	100%

New BTQ enters circulation through mining until the protocol-defined issuance is complete.

24. Mining Parameters

Mining secures the BitQube blockchain.

Algorithm

KawPoW

Hardware

GPU mining.

Mining Methods

- Solo Mining
- Pool Mining

Block Production

Miners compete to solve Proof-of-Work calculations. The first miner to produce a valid block broadcasts it to the network, where it is independently verified by full nodes. Valid blocks receive the protocol-defined block reward and any applicable transaction fees.

25. Network Parameters

The BitQube network operates as a decentralized peer-to-peer system.

Components

- Full Nodes
- Mining Nodes
- Wallet Clients
- Mining Pools
- Blockchain Explorer

Protocol Parameters

Parameter	Value
Peer Discovery	Protocol Value
Maximum Block Size	Protocol Value
Transaction Format	Protocol Value
Signature Scheme	Protocol Value
Network Identifier	Protocol Value

The official BitQube software defines the authoritative network parameters.

26. Security Considerations

BitQube is designed with the following security objectives:

Distributed Consensus

Consensus is achieved through independent validation by miners and full nodes.

Immutable Ledger

Confirmed blocks become part of a cryptographically linked blockchain.

Transaction Verification

Every transaction must satisfy protocol validation rules before inclusion in a block.

Open Verification

Any participant may independently verify blockchain data by operating a full node.

Network Integrity

The security of the blockchain depends on active participation by miners and node operators, together with correct implementation of the protocol rules. Users are encouraged to keep wallet software updated, protect private keys, and obtain software only from official project sources.

27. Software Releases

The BitQube software is developed through versioned releases. Each release may include:

- Bug fixes
- Performance improvements
- Security enhancements
- Documentation updates
- Feature additions

Where practical, release notes describe changes, upgrade instructions, and any protocol impacts.

28. Version History

Version	Status	Description
v1.0	Initial Release	Mainnet launch
Future Versions	Planned	Maintenance, security, and feature updates

> *Technical Statement: The BitQube protocol is the definitive source of truth for consensus rules, protocol parameters, and network behavior. If there is any difference between this white paper and the implemented protocol, the protocol implementation governs the operation of the network.*

29. Governance Philosophy

BitQube is designed as a decentralized, open blockchain network. Governance is intended to support the long-term stability, security, and transparency of the protocol while encouraging community participation.

Guiding Principles

- Transparency in development and communication.
- Security-first approach to protocol changes.
- Open participation from the community.
- Continuous improvement through testing and review.
- Long-term sustainability of the network.

As the ecosystem evolves, governance processes may also evolve to better support users, miners, node operators, developers, and ecosystem participants.

30. Community Participation

A decentralized blockchain depends on an active and engaged community. Community members may contribute by:

- Mining BTQ.
- Running full nodes.
- Testing new software releases.
- Reporting bugs and vulnerabilities responsibly.
- Improving documentation.
- Building applications and tools.
- Supporting community education and outreach.

31. Compliance Considerations

Blockchain networks operate in different legal and regulatory environments around the world.

Users, miners, businesses, and service providers are responsible for understanding and complying with the laws and regulations that apply in their jurisdictions.

The BitQube network itself is designed as decentralized blockchain infrastructure. However, organizations that build products or services around the network may have additional legal or regulatory obligations depending on their activities.

32. Risk Management

Participation in the BitQube ecosystem involves various risks.

Technical Risks

- Software bugs.
- Network interruptions.
- Hardware failures.
- Cybersecurity threats.

Market Risks

- Price volatility.
- Liquidity constraints.
- Changes in market demand.

Regulatory Risks

- Changes in digital asset regulations.
- Tax obligations.
- Licensing requirements for certain services.

User Risks

- Loss of wallet backups or private keys.
- Sending funds to incorrect addresses.
- Downloading unofficial software.

Users should evaluate these risks before participating in the network.

33. Future Vision

The long-term vision of BitQube is to provide reliable, decentralized blockchain infrastructure that supports innovation and practical applications.

Future areas of development may include:

- Continued improvements to network performance.
- Enhanced wallet functionality.
- Improved developer tools and documentation.
- Expanded ecosystem services.
- Greater community participation.
- Integration with additional applications and services.

34. Conclusion

BitQube is an independent Layer-1 Proof-of-Work blockchain secured by the KawPoW mining algorithm. The project is built around the principles of decentralization, transparency, security, and community participation.

By combining GPU-based mining, an open peer-to-peer network, and a predictable economic model, BitQube seeks to provide a stable foundation for its users and the broader ecosystem.

The continued success of the network depends on responsible development, active participation from miners and node operators, and ongoing collaboration with the community.

Acknowledgements

The BitQube project recognizes the contributions of developers, miners, node operators, researchers, testers, translators, partners, and community members whose participation helps strengthen the network.

Open collaboration and responsible innovation remain central to the continued development of the BitQube ecosystem.

35. Developer Ecosystem

BitQube encourages developers to build software, tools, and services that interact with the blockchain. A healthy developer ecosystem can contribute to network adoption, usability, and long-term sustainability.

Potential development areas include:

- Wallet applications
- Blockchain explorers
- Mining software integration
- Mining pool software
- Payment gateways
- Merchant tools
- Portfolio trackers
- Block explorers and analytics
- Mobile applications
- Open-source libraries

36. Exchange Integration

Digital asset exchanges considering BTQ integration should evaluate the network according to their own technical, legal, compliance, and risk management requirements.

A typical exchange integration includes:

- Running one or more full nodes.
- Synchronizing the blockchain.
- Generating deposit addresses.
- Monitoring blockchain confirmations.
- Processing deposits after the required number of confirmations.
- Supporting secure withdrawals.

Exchanges should independently verify protocol behavior and perform their own security testing before listing BTQ.

37. Merchant Integration

Businesses that choose to accept BTQ may integrate blockchain payments into their existing systems. A typical payment flow includes:

- Merchant generates a payment address.
- Customer sends BTQ.
- Merchant monitors the blockchain for confirmations.
- Payment is confirmed according to the merchant's chosen confirmation policy.
- Goods or services are delivered.

Each merchant should determine its own operational procedures, accounting practices, and compliance obligations.

38. Wallet Integration

Wallet software interacting with the BitQube blockchain should support, where appropriate:

- Address generation
- Transaction creation and signing
- Balance queries
- Transaction history
- Blockchain synchronization
- Backup and recovery
- Secure key management

39. Mining Pool Integration

Mining pools supporting BTQ should:

- Connect to synchronized full nodes.
- Distribute mining work to connected miners.
- Validate shares according to pool rules.
- Submit valid blocks to the BitQube network.
- Implement transparent reward accounting and payout policies.

40. API & Developer Resources

The BitQube ecosystem may provide developer resources such as:

- Node software
- Wallet software
- Technical documentation

- API references
- Source code repository
- Release notes

41. Responsible Disclosure

Security researchers and developers who identify potential vulnerabilities are encouraged to report them responsibly through the project's designated communication channels. Responsible disclosure helps improve software quality and protects the broader BitQube community.

42. Ecosystem & Adoption Strategy

Ecosystem Vision

BitQube is intended to provide a secure, decentralized Layer-1 blockchain that can support a growing ecosystem of users, developers, businesses, and infrastructure providers.

The long-term objective is to encourage practical use of the BTQ blockchain through open participation and community-driven development.

Ecosystem Participants

Miners

GPU miners secure the network by validating transactions and producing new blocks through the KawPoW Proof-of-Work consensus mechanism.

Node Operators

Full nodes independently verify protocol rules, maintain blockchain data, and contribute to network resilience.

Developers

Developers can create wallets, explorers, applications, integrations, and tools that interact with the BitQube blockchain.

Businesses

Businesses may choose to integrate BTQ as a payment option or incorporate the blockchain into products and services where appropriate.

Community

Community members contribute through education, testing, feedback, documentation, and participation in the network.

Adoption Strategy

BitQube seeks to encourage adoption through:

- Accessible documentation.
- Open participation.
- Community engagement.
- Reliable network infrastructure.
- Support for developers building on the ecosystem.
- Collaboration with businesses and service providers.

Long-Term Objectives

- Maintaining a secure and decentralized blockchain.
- Encouraging participation from miners and node operators.
- Supporting developers building tools and applications.
- Expanding ecosystem infrastructure.
- Improving documentation and user experience.
- Continuing protocol maintenance and security updates.

43. Future Research & Technical Development

BitQube is intended to evolve through ongoing research, software engineering, testing, and community feedback. Areas of research may include:

- Performance optimization
- Network scalability
- Node synchronization improvements
- Wallet usability enhancements
- Mining software compatibility
- Blockchain indexing and analytics
- Developer tooling
- Cross-chain interoperability, where appropriate

Software Development Principles

Security First

Security is a primary consideration for protocol changes and software releases.

Backward Compatibility

Where practical, changes should minimize unnecessary disruption for users, miners, and service providers.

Transparency

Development progress, release notes, and technical documentation should be communicated openly through official project channels.

Testing

New features and protocol changes should undergo appropriate testing before release.

Long-Term Sustainability

The continued operation of the BitQube network depends on several factors, including:

- Active participation by miners and node operators.
- Ongoing software maintenance.
- Secure infrastructure.
- Community engagement.
- Clear technical documentation.
- Responsible governance and development practices.

Appendix K — BitQube at a Glance

Category	Description
Blockchain	BitQube
Native Coin	BTQ
Blockchain Type	Layer-1
Consensus	Proof-of-Work
Mining Algorithm	KawPoW
Network	Peer-to-Peer
Ledger	Public Blockchain
Mining Hardware	GPU
Maximum Supply	8,000,000 BTQ
Premine	641,600 BTQ
Mineable Supply	7,358,400 BTQ

Appendix L — Block Lifecycle

Every BitQube block follows the same lifecycle:

- User creates a transaction.
- Transaction is digitally signed.
- Transaction is broadcast to the network.
- Full nodes verify the transaction.
- The transaction enters the memory pool.
- Miners select transactions.
- KawPoW calculations are performed.
- A miner discovers a valid block.
- The block is broadcast to the network.
- Full nodes validate the block.
- The blockchain is updated.
- Mining begins for the next block.

This process repeats continuously to maintain the blockchain.

Appendix M — Mining Lifecycle

The mining process consists of:

- Running compatible mining software.
- Connecting to a BitQube node or mining pool.
- Performing KawPoW computations.
- Discovering valid blocks or shares.
- Receiving rewards according to the protocol or pool payout method.

Mining contributes to both transaction validation and network security.

Appendix N — Security Best Practices

Participants are encouraged to:

- Download software only from official project sources.
- Verify software releases where possible.
- Keep operating systems and wallet software updated.
- Secure wallet backups and private keys.
- Use strong authentication for systems under their control.
- Monitor announcements for security updates.

Failure to protect wallet credentials may result in permanent loss of access to funds.

Appendix O — Frequently Asked Questions

What is BitQube?

BitQube is an independent Layer-1 Proof-of-Work blockchain secured by the KawPoW mining algorithm.

What is BTQ?

BTQ is the native coin of the BitQube blockchain.

Who can mine BTQ?

Anyone with compatible GPU hardware and mining software, subject to the laws and regulations applicable in their jurisdiction.

Does BitQube have a fixed supply?

Yes. The protocol defines a maximum supply of 8,000,000 BTQ.

How is the network secured?

The network is secured through decentralized Proof-of-Work mining and independent validation by full nodes.

Revision History

Version	Date	Description
1.0 Draft	Initial Release	First complete draft of the BitQube White Paper
Future Versions	TBD	Updates reflecting protocol improvements and ecosystem developments

Note: this Revision History table (from the white paper's Reference Guide appendix) describes v1.0 as the first complete draft of the document. Section 28's Version History table, provided separately in the source material, instead describes v1.0 as an initial mainnet release. Both are reproduced here as given, without reconciling the discrepancy.

44. Technical Standards & Operational Guidelines

Appendix F — Network Participation

BitQube is designed as an open and decentralized blockchain network. Participation is voluntary and may include mining, operating full nodes, developing software, providing infrastructure, or building services that interact with the network.

Participants are encouraged to:

- Operate current, supported software versions.
- Verify software obtained from official project sources.
- Keep systems updated with security patches.
- Follow documented installation and upgrade procedures.
- Report software issues through official communication channels.

Appendix G — Software Release Process

Development

New features and improvements are designed, implemented, and documented.

Testing

Changes should undergo functional, performance, and security testing before release.

Publication

Release packages should include version number, release date, changelog, upgrade instructions, and known issues.

Maintenance

After release, the project may issue updates that address defects, security concerns, or performance improvements.

Appendix H — Documentation Standards

Project documentation should aim to be accurate, clear, consistent, technically verifiable, and kept up to date as the protocol evolves.

Appendix I — Ecosystem Infrastructure

A mature blockchain ecosystem commonly relies on multiple independent infrastructure providers, including wallet providers, blockchain explorers, mining pools, node hosting services, developer tooling, analytics platforms, and community support channels.

Appendix J — Responsible Network Use

Participants are encouraged to protect private keys, run secure and updated software, verify transaction details before submission, respect applicable laws, and contribute constructive feedback.

45. Conclusion

BitQube (BTQ) is a decentralized Layer-1 Proof-of-Work blockchain designed to provide secure, transparent, and community-driven digital infrastructure through GPU mining. By utilizing the KawPoW consensus algorithm, BitQube promotes broad mining participation while maintaining network security and decentralization.

The BitQube ecosystem is intended to support real-world utility through infrastructure services such as GPU cloud mining, dedicated servers, hosting services, and future ecosystem applications. As adoption grows, additional utilities and integrations may be developed by the community and ecosystem participants.

The long-term vision of BitQube is to establish a sustainable blockchain ecosystem where miners, developers, businesses, and users contribute to the growth and security of the network.

Security Philosophy

- Decentralized Proof-of-Work security
- Community-operated mining
- Public blockchain verification
- Open-source software
- Transparent protocol rules
- Continuous security improvements

Open Source Development

BitQube supports transparent development through open-source software, released under the MIT license. Independent developers are encouraged to review, contribute, and improve the ecosystem.

Future Improvements

- Performance optimizations
- Wallet enhancements
- Developer tools
- Additional ecosystem applications
- Improved infrastructure
- Enhanced user experience

46. Glossary

Address — A destination used to send and receive BTQ.

ASIC — Application-Specific Integrated Circuit.

Block — A collection of validated transactions recorded on the blockchain.

Block Reward — Newly created BTQ issued to miners for successfully producing blocks.

Blockchain — A distributed ledger maintained by network participants.

Blocknomics — The economic model governing the issuance, distribution, and sustainability of BTQ.

Consensus — The process by which network participants agree on the valid state of the blockchain.

Full Node — A participant that stores and independently verifies the complete blockchain.

GPU — Graphics Processing Unit used for BitQube mining.

GPU Mining — Using graphics processing units to perform Proof-of-Work computations.

Hash Rate — The amount of computational work performed by miners.

KawPoW — The Proof-of-Work mining algorithm used by the BitQube network.

Layer-1 Blockchain — An independent blockchain with its own consensus mechanism and native coin.

Miner — A participant who contributes computational power to secure the network and produce blocks.

Node — Software that validates and relays blockchain data.

Peer-to-Peer Network — A decentralized network where nodes communicate directly.

Proof-of-Work — A consensus mechanism where miners perform computational work to validate transactions and create new blocks.

Wallet — Software used to manage cryptographic keys and interact with the blockchain.

47. References

Readers should refer to the official BitQube resources for the latest software releases, documentation, and network information.

- Official Website — https://bitqube.org
- Blockchain Explorer — https://explorer.bitqube.org
- Mining Pool — stratum+tcp://pool.bitqube.org
- Source Code Repository — https://github.com/bitqube/bitqube

Final Statement

BitQube represents a community-driven effort to build a secure and decentralized GPU-mined blockchain. Through transparent development, open participation, and practical infrastructure, the project aims to provide a robust foundation for future innovation. Continued collaboration among miners, developers, businesses, and users will be essential to the long-term success and sustainability of the BitQube ecosystem.

> *This white paper describes the intended design and objectives of the BitQube blockchain. As the protocol and ecosystem evolve, this document may be updated to reflect technical improvements, governance decisions, and ecosystem developments. Readers should always refer to the official BitQube software implementation and documentation as the authoritative source for protocol behavior and network operation.*